

EXPERIMENT 7

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Question 1:

Consider a relation R having attributes R(ABCD), functional dependencies: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Answer:

Closures:

- $B^+ = B$
- $AB^+ = ABCD$
- $BC^+ = BCDA$
- $BD^+ = BDAC$

Candidate Keys = {**AB, BC, BD**}

Prime Attributes = {**A, B, C, D**}

Non-Prime Attributes = {}

Since all RHS attributes are prime, the relation is in **3NF**.

For BCNF:

- $C \rightarrow D$, but C is not a superkey.
- Therefore, BCNF is violated.

Highest Normal Form = 3NF

Question 2:

Relation R(ABCDE) has functional dependencies: $A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R. Find the highest normal form.

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Answer:

Closures:

- $C^+ = C$
- $AC^+ = ACDBE$
- $BC^+ = BCAD E$
- $CD^+ = CD$
- $CE^+ = CE$

Candidate Keys = {**AC, BC**}

Prime Attributes = {**A, B, C**}

Non-Prime Attributes = {**D, E**}

Since $A \rightarrow D$ and A is a proper subset of candidate key AC, there is a **partial dependency** on a non-prime attribute.

Therefore, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 3:

Consider a relation R(ABCDE), with functional dependencies: $B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R. Find the highest normal form.

Answer:

Closures:

- $B^+ = BACDE$
- $A^+ = ACBED$
- $C^+ = C$
- $D^+ = D$
- $E^+ = E$

Candidate Keys = {**A, B**}

Prime Attributes = {**A, B**}

Non-Prime Attributes = {**C, D, E**}

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For BCNF:

- A and B are candidate keys, hence superkeys.
- All non-trivial FDs have a superkey on the left side.

Therefore, the relation satisfies **BCNF**.

Highest Normal Form = BCNF

Question 4:

Consider a relation R(ABCDEF), with functional dependencies: $A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$. Find the highest normal form.

Answer:

Since F never appears on the RHS of any FD, F must be present in every candidate key.

Closures:

- $AF^+ = ABCDEF$
- $BF^+ = ABCDEF$
- $DF^+ = ABCDEF$

Candidate Keys = {**AF, BF, DF**}

Prime Attributes = {**A, B, D, F**}

Non-Prime Attributes = {**C, E**}

Consider $A \rightarrow C$:

- A is a proper subset of candidate key AF.
- C is a non-prime attribute.
- Therefore, $A \rightarrow C$ is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 5:

Relation R(ABCDE) has dependencies: $CE \rightarrow D$, $D \rightarrow B$, $C \rightarrow A$. Find the highest normal form and candidate key set.

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Answer:

Closure:

- $CE^+ = CEDBA$

Candidate Key = {CE}

Prime Attributes = {C, E}

Non-Prime Attributes = {A, B, D}

Consider $C \rightarrow A$:

- C is a proper subset of candidate key CE.
- A is a non-prime attribute.
- Therefore, $C \rightarrow A$ is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 6:

Relation R(ABCDEF) has dependencies: $AB \rightarrow C$, $DC \rightarrow AE$, $E \rightarrow F$. Find the highest normal form and candidate key set.

Answer:

Since B and D never occur on the RHS, they must be part of every candidate key.

Closures:

- $BD^+ = BD$
- $BDA^+ = BDACEF$
- $BDC^+ = BDCAEF$

Candidate Keys = {ABD, BCD}

Prime Attributes = {A, B, C, D}

Non-Prime Attributes = {E, F}

Consider $DC \rightarrow E$:

- DC is a proper subset of candidate key BCD.
- E is a non-prime attribute.

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- Therefore, $DC \rightarrow E$ is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 7:

Relation R(ABCDEFGHI) has dependencies: $AB \rightarrow C$, $BD \rightarrow EF$, $AD \rightarrow GH$, $A \rightarrow I$. Find the highest normal form and candidate key set.

Answer:

Since A, B, and D never occur on the RHS, they must be part of every candidate key.

Closure:

- $ABD^+ = ABCDEFGHI$

Candidate Key = {**ABD**}

Prime Attributes = {**A, B, D**}

Non-Prime Attributes = {**C, E, F, G, H, I**}

Consider $AB \rightarrow C$:

- AB is a proper subset of candidate key ABD.
- C is a non-prime attribute.
- Therefore, $AB \rightarrow C$ is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 8:

Relation R(ABCDE) has dependencies: $AB \rightarrow CD$, $D \rightarrow A$, $BC \rightarrow DE$. Find the highest normal form and candidate key set.

Answer:

Since B does not appear on the RHS, B must be part of every candidate key.

Closures:

- $BC^+ = ABCDE$

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- $AB^+ = ABCDE$
- $BD^+ = ABCDE$

Candidate Keys = {**BC, AB, BD**}

Prime Attributes = {**A, B, C, D**}

Non-Prime Attributes = {**E**}

For 3NF:

- $AB \rightarrow CD$: AB is a superkey.
- $D \rightarrow A$: D is not a superkey, but A is a prime attribute.
- $BC \rightarrow DE$: BC is a superkey.

Therefore, the relation satisfies **3NF**.

For BCNF:

- $D \rightarrow A$, but D is not a superkey.
- Hence, BCNF is violated.

Highest Normal Form = 3NF

Question 9:

Relation R(ABCDE) has dependencies: $BC \rightarrow ADE$, $D \rightarrow B$. Find the highest normal form and candidate key set.

Answer:

Since C does not appear on the RHS, C must be part of every candidate key.

Closures:

- $BC^+ = ABCDE$
- $CD^+ = ABCDE$

Candidate Keys = {**BC, CD**}

Prime Attributes = {**B, C, D**}

Non-Prime Attributes = {**A, E**}

For 3NF:

- $BC \rightarrow ADE$: BC is a superkey.

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- $D \rightarrow B$: D is not a superkey, but B is a prime attribute.

Therefore, the relation satisfies **3NF**.

For BCNF:

- $D \rightarrow B$, but D is not a superkey.
- Hence, BCNF is violated.

Highest Normal Form = 3NF

Question 10:

Relation R(ABC) has dependencies: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$. Find the highest normal form and candidate key set.

Answer:

Closures:

- $A^+ = ABC$
- $B^+ = ABC$
- $C^+ = ABC$

Candidate Keys = {A, B, C}

Prime Attributes = {A, B, C}

Non-Prime Attributes = {}

A, B, and C are all candidate keys and therefore superkeys.

All non-trivial functional dependencies have a superkey on the LHS.

Therefore, the relation satisfies **BCNF**.

Highest Normal Form = BCNF

Question 11:

Consider a relation P having fields F1, F2, F3, F4, F5 with functional dependencies: $F1 \rightarrow F3$, $F2 \rightarrow F4$, $(F1, F2) \rightarrow F5$. In terms of normalization, which normal form is this table in? Describe the prime and non-prime attributes.

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Answer:

Since F1 and F2 never occur on the RHS, they must be part of the candidate key.

Closure:

- $F1F2^+ = F1F2F3F4F5$

Candidate Key = {**F1F2**}

Prime Attributes = {**F1, F2**}

Non-Prime Attributes = {**F3, F4, F5**}

Consider $F1 \rightarrow F3$:

- F1 is a proper subset of candidate key F1F2.
- F3 is a non-prime attribute.
- Therefore, it is a **partial dependency**.

Similarly, $F2 \rightarrow F4$ is also a partial dependency.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF